

Roll Number:

CSIC 103



NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
B.TECH., 1st SEMESTER, END-SEMESTER EXAM (AY 2025-26)
SUBJECT: Problem solving and Programming skills
COURSE CODE: CSIC 103

Maximum Marks: 50M

Time: 180 minutes

Answer all questions.

Q.1 For each of the following, write a C statements that performs the indicated task (snippet view only) — Do Not write the entire program. 5*2M

- A. Declare and initialize an integer variable *x* with any value and display its value.
- B. Declare a float variable *marks* read its value from the keyboard and display its value.
- C. Declare and initialize a character variable *grade* and display its value.
- D. Declare an integer array *arr* of size 5 read values into the array from the keyboard, and display the array values.

Q.2 **E.** Declare a character array *name* of size 20 read a string from the keyboard and display it. 2M
A. What is the role of the final *else* in an if-else-if ladder, and is it mandatory? Similarly, what is the role of the *default* case in a switch statement, and is it compulsory?

B. Rewrite the following code **without switch** construct.

And also determine the final values of *i* and *s* and illustrate the step-by-step execution when the input value of *i* is: (a) 2 (b) 4 (c) 5. 5M
3M

```
#include <stdio.h>
int main() {
    int i, s = 0;
    printf("Enter i: ");
    scanf("%d", &i);
    switch (i % 5) {
        case 4:
            s = i++;
        case 2:
            i++;
            break;
        case 0:
            s += 2 * i;
            i++;
            break;
        default:
            s = s + i;
    }
    printf("i=%d s=%d\n", i, s);
    return 0;
}
```

Roll Number: Q.3 A square matrix A of order $n \times n$ is given. The user has exactly 4 option matrices (M1, M2, M3, M4) that may contain the inverse of A. Using 2D arrays in C, write a program to: (i) Read the fixed matrix A. (ii) Ask the user to enter one option matrix at a time. (iii) For each entered matrix Mi, compute $C = A \times M_i$ (iv) Check whether C is an identity matrix (v) If the correct inverse is found, display which option (1-4) is the inverse and stop the program. If none of the four options is correct, display the message: "The inverse of the matrix is not present among the 4 options."	2M
Q.4 A. Mention the stages required for creating and using a user-defined function, and explain the different ways in which arguments can be passed to the function. B. What is the output of the following C program? Write the output and explain your answer by describing what happens at each statement execution. <pre>#include <stdio.h> int mystery (int ,int ,int); int main() { int x=1, y=2, z=0; z = mystery(x, y, z); printf("z=%d\n", z); if (z != 3) printf("STRANGE\n"); else printf("FUNNY\n"); } int mystery(int a, int b, int c) { c = a + b; return b - a; }</pre> C. Write an implementation of the user-defined C function <code>void crazy (char s[])</code>. Before the function call, s is a string terminated by a null character ('\0'). After the function has executed, all lowercase alphabetic characters in s should be replaced by #, all uppercase alphabetic characters should be replaced by &, and all remaining characters should be retained as they are.	5M
Q.5 A. Define a structure named student with the following members: Name of size 20 bytes, Registration Number, Branch, CGPA B. Write a C program to: (i) Accept the number of students n from the user. (ii) Use only one structure variable to take inputs for each student at a time. (iii) Store the data of all n students on the hard disk in a file named <code>studentrecord.dat</code>.	2M 8M